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# Video 3 · Algebra II Worked Solutions

Cambridge IGCSE Mathematics 0580, Paper 2 Non-calculator (Extended). Full worked solutions to Questions 12–15 of the NurNabi Academy Full Revision Paper, solved the simplest way, with the method and accuracy marks shown at every step, the way a Cambridge examiner awards them.

| 4 | 16 | Q12–Q15 | 16:50 |
|---|---|---|---|
| questions | marks | E2.7 – E2.13 | video runtime |

M1 method mark  A1 accuracy mark (needs the M mark)  B1 independent mark

## Question 12

# Linear sequences

**Find the *n*th term of 5, 12, 19, 26, …** [2]

| 5 | 12 | 19 | 26 |
|---|---|---|---|
| | +7 | +7 | +7 |

The difference is 7, so the rule starts with **7*n***. B1

Compare with the 7 times table: 7, 14, 21, 28 …

Each term is 2 less, so subtract 2.

***7n − 2*** B2

**Marks:** B2 for $7n - 2$. B1 only for $7n + k$ with the wrong $k$.

**Quick way to get the number:** the term before the 1st would be $5 - 7 = -2$. That's the zeroth term, the number after $7n$.

**Check** $n = 3$: $21 - 2 = 19$ ✓. Common mistake: writing $n + 7$.

## Question 13

# Direct proportion

$y$ is directly proportional to $\sqrt{x}$. When $x = 9$, $y = 12$. Find $y$ when $x = 25$. [3]

**1. Write the equation with $k$:** $y = k\sqrt{x}$ M1

**2. Find $k$:** $12 = k \times \sqrt{9} = 3k \rightarrow k = 4$ A1

**3. Use it:** $y = 4 \times \sqrt{25} = 4 \times 5$

**$y = 20$** A1

**Always write the full formula** $y = 4\sqrt{x}$ before using it. If a later part asks for $x$ given $y$, you'll need it.

**Common mistake:** using $y = kx$ (forgetting the square root), which gives $k = \frac{4}{3}$ and $y = 33.3\ldots$

## Question 14

# Functions — composite and inverse

$f(x) = 2x - 3 \quad g(x) = \frac{1}{x+1}$

**(a) Find f(4).** [1]

Replace $x$ with 4: $f(4) = 2 \times 4 - 3$

**5** B1

**(b) Find gf(3).** [1]

**gf means do f first**, then g (work from the inside out).

$f(3) = 2 \times 3 - 3 = 3$

$g(3) = \frac{1}{3+1}$

**$\frac{1}{4}$** B1

**Trap:** doing g first gives $f(g(3)) = -2.5$, which is wrong.

**Question 14** (continued)

# Functions — composite and inverse

**(c) Find $f^{-1}(x)$.** [2]

Write $y = 2x - 3$, then **swap** $x$ and $y$:

$x = 2y - 3$ M1

Rearrange: $x + 3 = 2y \rightarrow y = \frac{x+3}{2}$

$$f^{-1}(x) = \frac{x+3}{2}$$ A1

**Check:** f(4) = 5, and $f^{-1}(5) = 8 \div 2 = 4$ ✓

**(d) Solve ff($x$) = 11.** [2]

Put f($x$) into f:

$ff(x) = 2(2x - 3) - 3 = 4x - 9$ M1

$4x - 9 = 11 \rightarrow 4x = 20$

$$x = 5$$ A1

**Check:** f(5) = 7, f(7) = 11 ✓. ff($x$) is **not** f($x$) × f($x$).

## Question 15

# Differentiation and turning points

$y = x^3 - 6x^2 + 9x + 2$

**(a) Find $\frac{dy}{dx}$.** [1]

Multiply by the power, then reduce the power by 1:

```
x³  → 3x²
6x² → 12x
9x  → 9
2   → 0
```

$$\frac{dy}{dx} = 3x^2 - 12x + 9$$ **B1**

**(b) Find the coordinates of the two turning points.** [3]

At a turning point the gradient is 0:

$3x^2 - 12x + 9 = 0$ **M1**

$\div 3$: $x^2 - 4x + 3 = 0 \rightarrow (x - 1)(x - 3) = 0$

$x = 1$ or $x = 3$ **A1**

Substitute into the **original** equation:

```
x = 1: 1 - 6 + 9 + 2   = 6
x = 3: 27 - 54 + 27 + 2 = 2
```

**(1, 6) and (3, 2)** **A1**

**Question 15** (continued)

# Differentiation and turning points

**(c) Determine which turning point is a maximum. Show how you decide.** [1]

Differentiate again: $\frac{d^2y}{dx^2} = 6x - 12$

At $x = 1$: $6 - 12 = -6$, which is **negative**, so a **maximum**

At $x = 3$: $18 - 12 = 6$, which is positive, so a minimum

**(1, 6) is the maximum** B1

**Memory trick:** negative second derivative = a "sad face" ∩ = maximum. Positive = "happy face" ∪ = minimum.

**Other valid reason** (easier for some students): the curve is a positive cubic, so it goes up, turns down, then turns up. The **first** turning point (smaller $x$) is the maximum. Also, $y = 6 > y = 2$.

**The mark needs a reason.** "(1, 6)" alone scores 0.

## Answers at a glance

# Video 3 · Algebra II · 16 marks

| Q | Topic | Answers | Marks |
|---|---|---|---|
| 12 | Linear sequences | $7n - 2$ | 2 |
| 13 | Direct proportion | $y = 20$ | 3 |
| 14 | Functions | (a) 5 (b) $\frac{1}{4}$ (c) $\frac{x+3}{2}$ (d) $x = 5$ | 6 |
| 15 | Differentiation | (a) $3x^2 - 12x + 9$ (b) (1, 6), (3, 2) (c) (1, 6) is the maximum | 5 |
| | **Total** | | **16** |

Mark allocations follow the style of the Cambridge 0580 mark scheme. The questions are original NurNabi Academy practice questions, not a prediction of the real paper. Next: **Video 4 · Coordinate geometry and geometry (Q16–Q19)**.